

The problem

Indian small-business owners have data, but not operating system for acting on it.

A merchant can see sales, customers and inventory, yet still cannot answer: Why did sales fall? Which customers should I reach? Is the offer affordable? Did it actually work? Most tools stop at charts or generic recommendations.

A typical merchant question

"Why were sales lower yesterday, and what should I do next?"

Language and voice first

Evidence, not guesswork

What Vyapaar AI changes

1

Understands the signal

Finds sales anomalies, lapsed regulars, stock risks and peer gaps from the merchant ledger.

2

Recommends a bounded action

Builds a costed campaign using grounded tools, guardrails and a measurable success condition.

3

Closes the loop

Requires approval, records the action, then measures an outcome against a control group.

The product promise

SENSE > DIAGNOSE > PRESCRIBE > APPROVE

EXECUTE > MEASURE > LEARN

The merchant stays in control. The system makes decisions explainable, approval-gated and traceable.

One merchant question becomes an accountable growth loop.

01

Sense

Scheduled detectors scan the transaction ledger for day-of-week dips, customer churn risk, stock signals and peer gaps.

02

Diagnose

The agent calls allowlisted data tools to retrieve sales, customers, inventory and historical context. It does not invent business numbers.

03

Prescribe

A playbook estimates audience, spend cap, expected impact and success criteria using code-based estimators.

04

Approve

A signed merchant approval is required before any customer-facing action. Policy code checks audience, frequency, spend and quiet hours.

05

Execute and measure

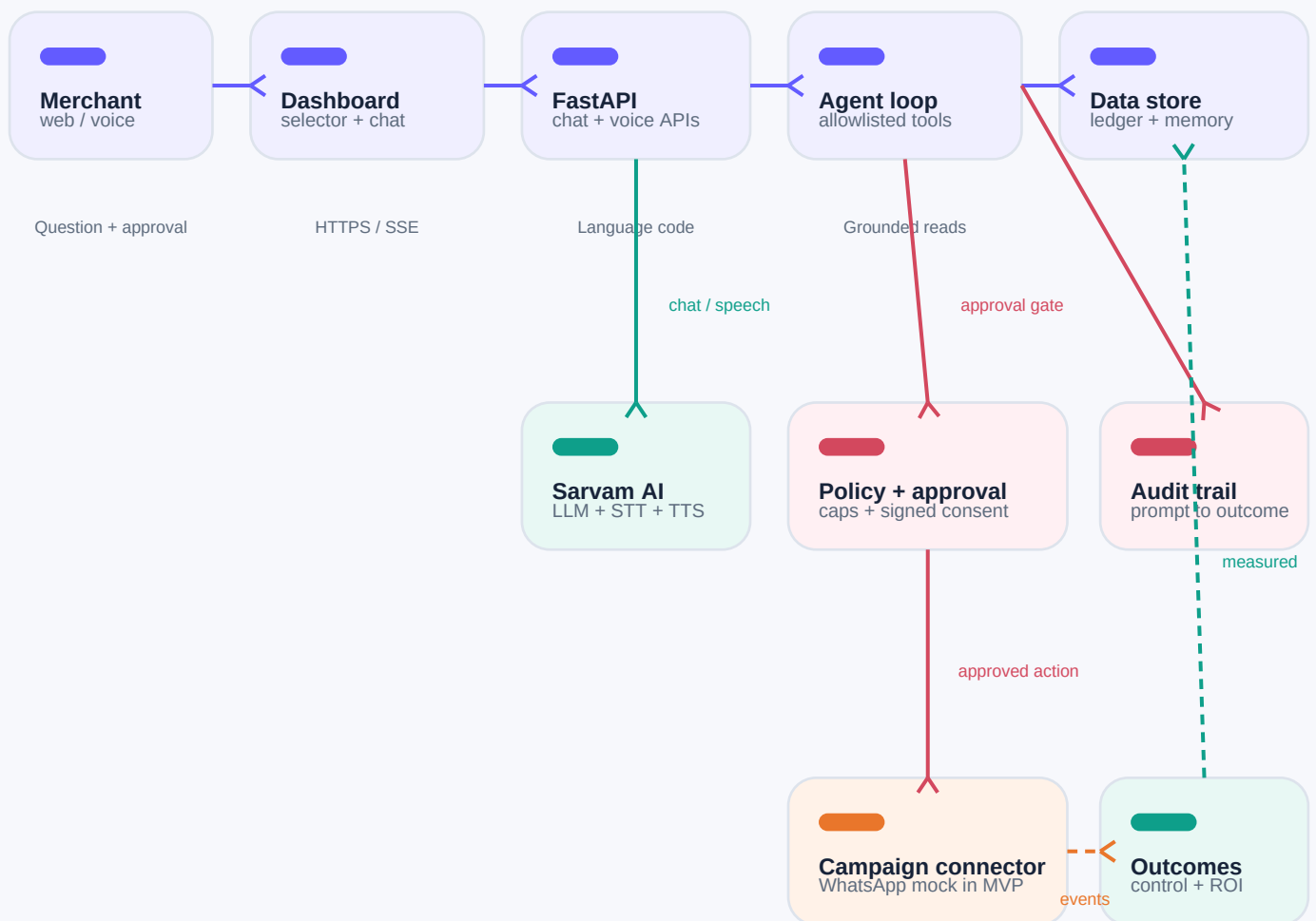
A campaign connector records delivery and redemption events. Outcome logic compares the treated group with matched never-messaged customers.

Language and voice are part of the workflow, not an afterthought.

The dashboard sends the selected language code with chat, speech-to-text and text-to-speech requests. Sarvam services provide multilingual AI and voice where configured; browser voice is the fallback. The UI supports English plus Hindi, Bengali, Tamil, Telugu, Kannada, Malayalam, Marathi, Gujarati, Punjabi and Odia.

Architecture: grounded AI with a human control point

Every component serves an explicit responsibility in the decision-to-action loop.



Design principles

- Grounded**
 Business facts are retrieved by tools from merchant data.
- Controlled**
 Policy and signed approval are code gates, outside the LLM prompt.
- Auditable**
 Questions, tools, decisions and outcomes are retained as a trace.
- Measured**
 Campaign results use a matched control group, not only raw sales lift.

Why this is more than a chatbot demo

Vyapaar AI is designed to turn recommendations into accountable operating decisions.



Detect with code

Sales anomalies and churn signals are produced by deterministic detectors with fallbacks.



Ground every number

The agent uses registered tools and a code-based impact estimator for figures and projections.



Keep the merchant in control

No outreach occurs before signed approval. Cost, frequency and quiet-hour limits are enforced in policy code.



Learn from impact

Delivery, redemption and financial outcome data feed back into the next decision.

Demo scenario

What the prototype can demonstrate

-18%

Sunday sales dip detected

212

lapsed regulars identified

Rs 4.8k-6.5k

expected GMV from estimator

4.84x

illustrative measured ROI

These are seeded, reproducible demonstration metrics. They are not claims about a live merchant deployment.

MVP status and credible next step

Working now

Sharma Tea Stall is the default demo. Import a sales CSV to replace the active workspace; chat, signals and forecasts then use your ledger. Restore the demo anytime.

Production path

Move durable merchant imports from temporary SQLite to Postgres and secure object storage; replace the mock connector with an approved WhatsApp BSP; add authentication and consent management.

Live app: vyapaar-ai-phi.vercel.app

Source: github.com/SunnyKumar28/Vyapar_AI